

Capturing traffic over Wi-Fi Voice Commands Privacy Analysis of Augmented Reality Glasses

1 Introduction

This experiment captures network traffic over Wi-Fi during the interaction between an Android phone and the Ray Ban Meta glasses, more specifically while executing voice commands. The aim of this data collection is to observe the network traffic generated during typical usage of the glasses for voice commands (to see how data flows between the device, its companion app and other apps). The goal would be to identify potential vulnerabilities or privacy breaches if data transmission is not securely protected. Furthermore, we can take a peek into whether any personally identifiable information or metadata can be taken during daily usage, therefore observe how user privacy may be affected.

2 Metadata

Time of experiment: Oct 16th 14:37

Duration of experiment: Roughly 50 minutes

IP addresses:

- Laptop - 192.168.0.215
- Redmi phone - 192.168.1.141
- Wi-Fi Network - 192.168.0.1
- ALFA Hotspot - 192.168.1.100/24
- Meta CDN (content delivery network) - 157.240.17.17

3 Voice Commands used

- "Hey Meta, what's the weather in [place] today?"
- "Hey Meta, what time is it?"
- "Hey Meta, what date is today?"
- "Hey Meta, what's the battery percentage?"

- "Hey Meta, play music."
- "Hey Meta, increase the volume."
- "Hey Meta, lower the volume."
- "Hey Meta, pause music."

- "Hey Meta, take a photo."
- "Hey Meta, take a video."

- "Hey Meta, stop recording."
- "Hey Meta, send a message to [name]."
- "Hey Meta, send a voice message to [name]."
- "Hey Meta, read my messages."
- "Hey Meta, set a timer for 1 minute."
- "Hey Meta, how much time is left in my timer."
- "Hey Meta, cancel my timer for 1 minute."

4 Time Stamps

5 Note

All packets: 34685

IPv4 - 97.5%

UDP - 8.6%

TCP - 88.8%

TLS - 35.7%

IPv4 -

10.42.0.34 to 157.240.17.17 (owned by Meta CDN), 28706

10.42.0.34 to 157.240.17.3 (owned by Meta), 1584

10.42.0.34 to 224.0.0.251 (Multicast DNS address), 748

10.42.0.34 to 10.42.0.1 (My router), 436

TCP -

10.42.0.34 to 157.240.17.17 (owned by Meta CDN), most

10.42.0.34 to 157.240.17.3 (owned by Meta), mostly the rest

UDP -

10.42.0.34 to 224.0.0.251 (Multicast DNS address), mostly

10.42.0.34 to 157.240.17.3 (owned by Meta), half

10.42.0.34 to 10.42.0.1 (My router), some